

The Instant, Memory and Resonance: Revisiting Lived Time

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Abstract

This article revisits lived time as a phenomenon of *local* coherence supported by the Chronon Field, $\Phi(x)$, defined as a background pulsation with dimension T^{-1} . We set the operational definition $T(x) = 1/\Phi(x)$ and model the present as an episode of *phase locking* between neural dynamics and this pulsation. Perception, attention and memory operate within finite *coherence windows* of order $\tau_c \sim 30\text{--}300$ ms, transiently aligned with $\Phi(x)$. Memory is understood as *phase re-entry*, attention as the capture of dominant frequencies within a stabilized tempo. We specify quantitative markers (bands 4–80 Hz, delays 1–20 ms, tracking of speech envelope 2–8 Hz) and relate them to *public time* (GNSS, PTP, NTP) as a *shared backbone*. The framework unifies phenomenology and physics under a logic of *beating*: coherence windows are born, maintained and closed on the pulsation of $\Phi(x)$, with a loss term $\Gamma(x)$ that governs phase holding.

Keywords: Chronon Field, lived time, resonance, synchronization, memory, attention, neural rhythms, public time

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I — Inner measurement

Time is not only read on dials: it is felt. It beats in the body, in breathing, in attention that stretches or contracts. Each experience of the present is a coherence window where perception, memory and action overlap without merging, with psychophysical thickness $\tau_c \sim 30\text{--}300$ ms, consistent with perceptual latency, simple decision times and the attentional blink around 200–500 ms.

Neurophysiology observes perception in temporal packets: 30–80 Hz (gamma), 13–30 Hz (beta), 8–12 Hz (alpha), 4–8 Hz (theta), depending on the task and vigilance state. Vision is segmented with a critical flicker fusion around $\sim 50\text{--}60$ Hz, audition discriminates gaps of a few milliseconds, and saccades rhythm sampling at $\sim 3\text{--}5$ Hz. These frequencies are *rhythms of presence*.

What do these rhythms rhythm? What maintains the fleeting stability where an instant takes shape? We posit the Chronon Field, $\Phi(x)$, as a non-neuronal physical pulsation that supports the cadence of phenomena from quantum to cognitive scales, and provides the locking backbone of coherence windows. We call *local beat* $T(x)$ the inverse of the pulsation, $T(x) = 1/\Phi(x)$; the lived instant corresponds to $T \approx \tau_c$ when the phase-compatibility condition is satisfied.

Box — Conventions & anchors (operational).

$$\boxed{d\tau = \Phi^{-1}(x) dt}, \quad \boxed{J^\mu = \Phi u^\mu}, \quad \boxed{\partial_\mu J^\mu = \Gamma(x)}.$$

Separation of loss terms: $\Gamma(x) = \Gamma_{\text{env}}(x) + \xi \Phi(x) + b |\nabla \Phi(x)|$ (dimensionless couplings ξ, b). Weak-field anchor (public time cartography): $\Delta f/f \simeq 1,1 \times 10^{-16} \text{ m}^{-1} \Delta h$. *Discipline*: Φ does not modify light cones nor the GR metric; it re-parameterizes *cadences*.

II — The present as a coherence beat

Each sensorimotor network has its own tempo: audition synchronizes microphases on 1–10 ms, working memory anchors in $f_\theta \approx 4\text{--}8\text{ Hz}$, vision alternates fast processing and slower integrations. Conduction velocities $v \in [1, 100]\text{ m s}^{-1}$ induce phase lags 1–20 ms over cortical distances of a few centimeters: coherence is a dynamical achievement.

The field $\Phi(x)$ provides the underlying rhythmic frame. The lived instant arises when neural dynamics briefly align with the field pulsation, satisfying a locking criterion

$$|\Delta\phi| < \varepsilon \quad \text{during} \quad T \approx \tau_c,$$

with a geometric-conductive phase lag

$$\Delta\phi \simeq 2\pi f \frac{d}{v}.$$

Examples: $d = 3\text{ cm}$, $v = 50\text{ m s}^{-1}$, $f = 40\text{ Hz} \Rightarrow \Delta t \simeq 0,6\text{ ms}$ and $\Delta\phi \approx 0,151\text{ rad}$; $d = 10\text{ cm}$, $v = 10\text{ m s}^{-1}$, $f = 8\text{ Hz} \Rightarrow \Delta t \simeq 10\text{ ms}$ and $\Delta\phi \approx 0,503\text{ rad}$. When alignment breaks down, one window closes and another opens. Theta–gamma coupling at encoding, motor beta rebounds, gamma bursts under focused attention can be interpreted as episodes of tuning to the backbone imposed by $\Phi(x)$. EEG/MEG oscillations are *not* $\Phi(x)$: they *align* with it when phase compatibility holds.

Box — Measurements & metrics. Alignment is assessed by: (i) *Phase Locking Value* (PLV) and ITPC; (ii) cross-spectral coherence $C_{xy}(f)$; (iii) circular phase variance. Hypotheses: **H1**: $\text{PLV}(f)$ increases when phase variability over T decreases (stability of Φ); **H2**: jitter Δt crossing $2\pi f \Delta t > \varepsilon$ degrades ITPC and performance. Role of loss: small $\Gamma \Rightarrow$ stable phase holding & robust memory; large $\Gamma \Rightarrow$ faster closure of windows.

III — Memory and phase persistence

Memory is not a deposit, but a continuity of resonance. A trace survives because it can retrieve the tempo of its appearance: to remember is to restore the internal frequency that made the event hold together. Converging observations show a reactivation of oscillatory spectra close to those at encoding (hippocampal theta, perceptual gamma), with a phase readjustment on the scale of $\sim 100\text{ ms}$ correlated with recall precision. In our framework, stability depends on a loss term $\Gamma(x)$: when the phase variability of $\Phi(x)$ is low (controlled temporal gradients), Γ decreases and the *reinstatement window* widens, with $\tau_{\text{reprise}} \gg \tau_c$ as soon as context restores phase compatibility.

IV — Resonance and attention

Attention acts like a tuning fork: it captures the dominant phase of a scene and lowers internal noise. In audition, alignment to 2–8 Hz speech envelopes increases intelligibility of the vocal stream; in vision, performance fluctuates at $\sim 7\text{--}10\text{ Hz}$ as if attentional sampling were scanning the scene in quasi-discrete steps. This ability presupposes a stable background: $\Phi(x)$ guarantees tempo continuity despite environmental fluctuations. The stream of consciousness becomes a succession of coherence windows relayed by the field pulsation; the “elasticity” of felt time—dilation, urgency—reflects coupling regimes where τ_c stretches or contracts with phase mismatch.

V — The self as rhythmic interface

The *self* appears as a node of coherence: a bundle of rhythms that remain in phase. In deep sleep (delta 0,5–4 Hz), trance, or some dementias, phase architecture loosens and the self dissolves; under sustained concentration, the rhythmic backbone tightens and presence consolidates.

Public time provides a shared standard — GNSS (ns), PTP/IEEE 1588 (sub- μs), NTP (ms) — while *lived time* measures the effective locking of our rhythms to $\Phi(x)$. The two meet when action requires fine coordination: ensemble music with a metronome, an electrical grid regulated at 50 Hz, surgery operating within $\sim 100\text{ ms}$ windows where movement and perception coincide.

VI — EEG/MEG experimental program (OSF preregistration)

Goal: test temporal coupling between neural rhythms and $\Phi(x)$ within windows $\tau_c = 30\text{--}300$ ms.

General protocol. – Multi-site EEG/MEG recordings, PTP synchronization (sub- μ s). – Rhythmic manipulations: modulation of input tempo (2–10 Hz) or of attention. – Analyses: PLV, wPLI, ITPC, cross-spectral coherence $C_{xy}(f)$. – OSF preregistration: dependent variables, filters, bands, windows, multiple-comparison corrections (FDR/cluster). – Expected effect size: $\Delta\text{PLV} \simeq 0,05\text{--}0,1$ between synchronous and desynchronized blocks. – Null blocks (conditions $\Delta t = 0$) included to test for spurious effects. – Increased vigilance on artifacts: volume conduction, muscle activity, I/O latencies.

VII — Risk of artifacts

Phase-coherence measures are sensitive to several major artifacts:

- **Volume conduction:** spurious couplings between nearby regions; mitigated by source-level reconstruction.
- **Muscle activity:** frontal or temporal contamination; dedicated EMG filtering and ICA-based detection.
- **I/O timing:** misalignment of stimulus and recording; correction by independent time-locking.
- **Null blocks:** blocks without modulation (synchronized baseline) to estimate the false positive rate of locking.

These precautions help ensure that changes in PLV/wPLI genuinely reflect synchronization effects to $\Phi(x)$ rather than artifactual correlations.

VIII — Conclusion: consciousness as a beat of the real

Perception is not a reflection, but a participation; memory is a holding, not a storage; the present is a local resonance of the field $\Phi(x)$. Brain, matter and world share a common breath: a pulsation. Far from being a spectator, the mind is a rhythm of the real upon itself. Consciousness is a beat of the world recognizing itself; and knowing, perhaps, is only this: learning how to keep the measure of the real.

Outlook: this article prepares the transition to an *Ontology of Rhythm* — a post-classical metaphysics in which $\Phi(x)$ becomes the load-bearing structure of the real.

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